Al-Balqa' Applied University
Faculty Of Engineering Technology
Mechatronics Engineering Dept.

Process Control
(3 )Class Hours 40 Minutes.)
Thirsday-9—12-2021
(12.00-13.00)
Room 1226

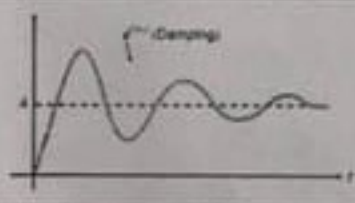
*Instructor ;Eng Monther Kan"an*

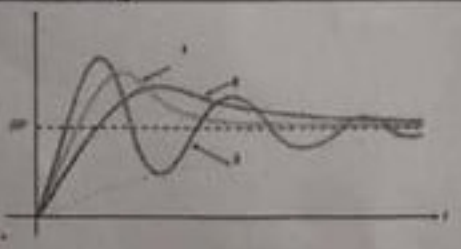
Mid term
30 Points

18/30

Student Name: Malek Maher Abu Ghazaleh.....
3191501803 8

**Q.1: Choose the correct answers ...** **{10Points}**

| | |
|---|---|
| 1. | *Proportional band of the controller is expressed as:* |
| | *a. Gain b. ratio c. range of control variable d. percentage* |
| 2. | *An element that translates the controller output into an action on the final element is called .....* |
| | *(a) Actuator (b) process (c) controller (d) feedback* |
| 3. | *The ..... computes the error signal and uses it to produce the control action.* |
| | *(a) controller (b) process (c) feedback (d) final element* |
| 4. | *The controller is called ON/ OFF controller, because the final element has one states on or off.....* |
| | *(a)True (b) False* |
| 5. | *The objective of control system is to make the process variable exactly zero.....* |
| | *(a)True (b) False* |
| 6. |  *1- In this figure This response is for* |
| | *a- Overdamped b- critically damped c- undamped d- Under damped* |
| 7. | *When you wait the temperature to reach to the set point this is called ?* |
| | *(a)dead time (b) lag time (c) dead band* |
| 8. | *Manipulated variable is the ..... That acted on by the controller .* |
| | *(a)controlled variable (b) control variable (c) desired value (d) process variable* |
| 9. | *If the error becomes more positive as the control variable increases the action of controller is called.....* |
| | *(a)Direct Acting (b) Reverse Acting* |

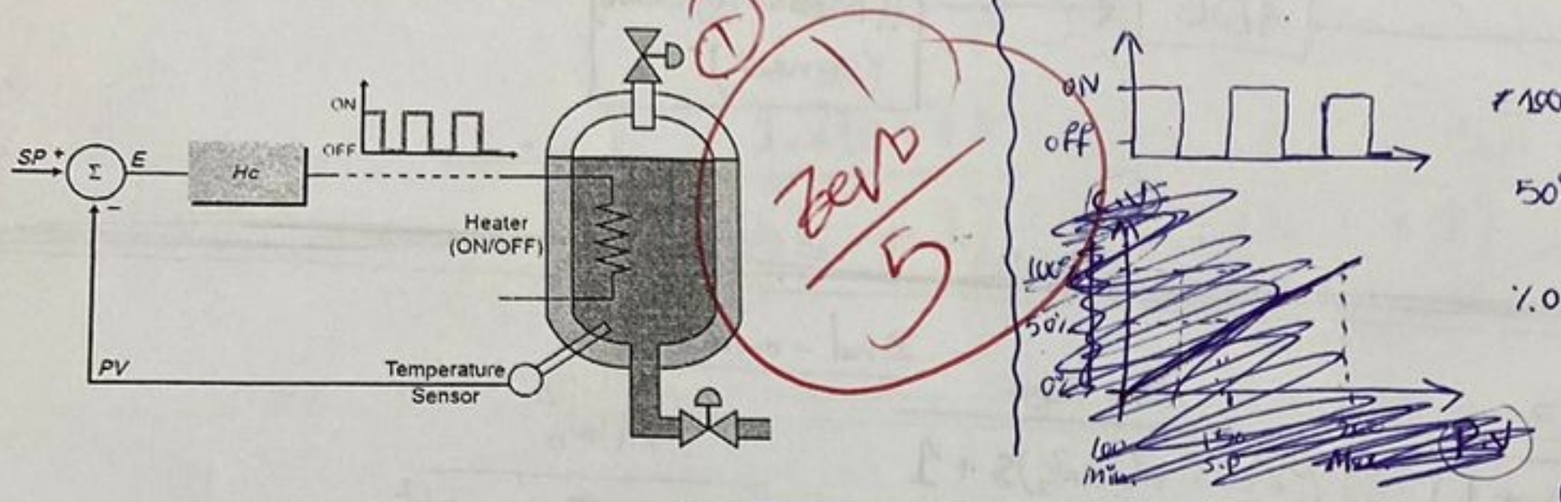


*10- In this figure .*

a-(1) over damped (2) critically damped (3) under damped
b-(1) over damped (3) critically damped (2) under damped
c-(3) over damped (2) critically damped (1) under damped
d- (2) over damped (1) critically damped (3) under damped

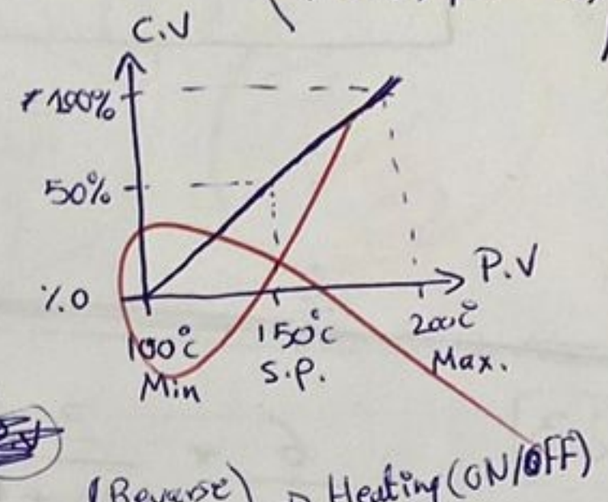
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| D | A | A | A | B | D | B | D | A | D |

**Q2:- Plot the response of control variable in the figure below ?** (6)







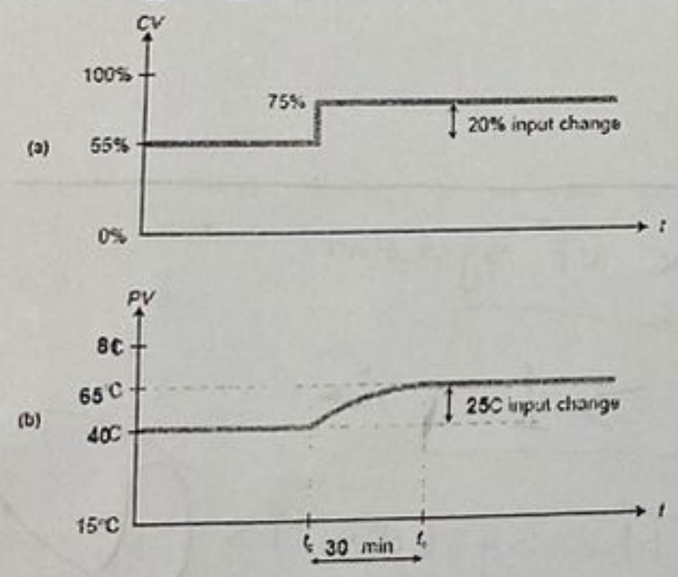



(Reverse Acting) → Heating (ON/OFF)

$$\frac{6}{5S+1} \cdot \frac{3}{2S+1}$$

**Q3- For this system find** (9)

**1-Ksys ,2- Wn ,3- $T_{sys}$ ,4- damping ratio ( £ )?,5- type of system ?**




**Q4- For the figure above**

a- Calculate the amount of process gain?

b- Write the process's transfer function for this system?